PHASE 1

A WLAN (Wireless Local Area Network) is a technology that allows devices like laptops, and smartphones, to connect to a local network and the internet without physical cables, typically using radio waves within a home, office, or campus. It operates over short distances, often using wifi for communication via access points.

- **Components: Consists of wireless access points (APs) or routers connecting devices to a wired network.**
- **Benefits: Provides mobility, flexibility, and easier, often cheaper installation compared to wired Ethernet networks.**

❖ **Differences Between LAN and WLAN:**

- **Connectivity: LAN uses physical cables (Ethernet), while WLAN uses wireless radio waves (Wi-Fi).**

- **Security: Wired LANs are generally more secure because they require physical access; WLANs are more vulnerable to interception.**
- **Installation: LANs require cable management, while WLANs are easier to set up but may require repeaters to extend range.**
- **A wireless access point (WAP or AP) is a networking hardware device that allows Wi-Fi enabled devices to connect to a wired network, typically via an Ethernet cable, to provide wireless internet access. It acts as a bridge between wired and wireless**

❖ <u>Key Facts About SSIDs</u>**:**

- **Purpose: Distinguishes one network from another in areas with multiple access points.**
- **Visibility: Routers typically broadcast their SSID, making them visible when scanning for networks on your phone or computer.**
- **Security: While SSID hides nothing on its own, it is used to identify the network, while WPA2/WPA3 passwords secure it.**

❖<u>WIRELESS ACCESS POINT</u>

Definition and How It Works

A wireless access point is a specialized hardware device that creates a wireless local area network (WLAN). Unlike a typical home router, its function is singular: to act as a bridge, extending a wired network infrastructure into the wireless domain so that Wi-Fi-enabled devices

can connect. The fundamental purpose of a wireless access point is to convert data from a wired Ethernet connection into wireless radio signals and vice versa. It broadcasts one or more Service Set Identifiers (SSIDs), which are the public names of the Wi-Fi networks that appear on your devices.

WHAT IS WPA2/WPA3?

WPA2

WPA2 has been a central feature of Wi-Fi security for many years, offering strong encryption and authentication methods to protect wireless networks. It relies on the Advanced Encryption Standard (AES) for secure data transmission and utilizes a specific handshake process to establish trusted connections between devices and access points. Despite its effectiveness, WPA2 has experienced vulnerabilities such as the KRACK exploit, highlighting the necessity for a more resilient security protocol

- **WPA3**

It uses special encryption for information that keeps it safe and private. Each tool and connection gets its own unique key to lock up data.

This stops hackers from getting your personal information.

WPA3 utilizes SAE, a stronger authentication method compared to the 4-way handshake used in WPA2, adding an extra layer of security against unauthorized access.

Enhanced Resistance to Modern Attacks

WPA3 gives better security from current dangers and weaknesses, especially those that were very well exploited in WPA2, like the KRACK attack.

❖ DIFFRENCES BETWEEN 2.4GHZZ AND 5 GHZ

- **The 2.4 GHz Wi-Fi band covers more area but at slower speeds, while 5 GHz provides higher speeds with a shorter range.**

- **For tasks requiring high bandwidth close to the router, like streaming or gaming, use 5 GHz; for broader coverage, 2.4 GHz is preferable.**